EMMA REY SÁNCHEZ

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SUMMARY

AI Engineer at Deloitte with a BSc in Mathematical Engineering with Artificial Intelligence, currently pursuing an MSc in Advanced Artificial Intelligence. Mathematical background in probability, optimisation and information geometry, with practical experience in NLP, computer vision, generative AI and model evaluation.

EXPERIENCE

Artificial Intelligence Engineer **September 2026 - Present**
Deloitte Barcelona, Spain

- Developed a Retrieval-Augmented Generation (RAG) system for context-informed LLM responses.
- Contributed to computer vision and AI agent projects for the pharmaceutical industry.

EDUCATION

MSc in Advanced Artificial Intelligence **Sep 2025 - Dec 2026 (expected)**
Universidad Pontificia Comillas ICAI Madrid, Spain

- Relevant modules: Deep Generative Models, Information Geometry, Probabilistic AI, Deep Reinforcement Learning.

Academic Exchange **January 2025 - June 2025**
University of Miami Miami, USA

- Engineering and AI, with business coursework. Grade: 9.4/10.

BSc in Mathematical Engineering with Artificial Intelligence **Sep 2021 - Jun 2025**
Universidad Pontificia Comillas ICAI Madrid, Spain

- Relevant modules: NLP, Deep Learning, Machine Learning, Computer Vision, Robotics and Reinforcement Learning.

A-Levels and GCSEs | Kew House School, London | 2016-2021
A-Levels: A*, A*, A*, A*, A* (Maths, Further Maths, Physics, Chemistry, Spanish).

COMPLEMENTARY EDUCATION

Diplomas in Personal and Professional Skills; Communication Skills and Foreign Language Studies
Universidad Pontificia Comillas

Sailing and Windsurfing Instructor Course | Escuela de Vela Socaire | June 2021

KEY ACHIEVEMENTS

Bachelor's Thesis: Hierarchical Music Genre Classification
Improved deep learning accuracy on the FMA dataset using two-stage training with macro-genres and NLP-based music metadata enhancement.

GNN Rewiring Analysis: GitHub
Compared four graph-rewiring strategies across three datasets, analysing over-squashing and over-smoothing with GAT models and graph diagnostics.

Bayesian Sentiment Analysis: GitHub
Co-developed a sentiment classifier using Bayesian mixture-of-experts, topic modelling and uncertainty estimation; evaluated calibration and out-of-distribution detection.

Image Captioning Project: GitHub
Implemented an InceptionV3-LSTM model with greedy and beam-search decoding, evaluated with BLEU and CIDEr.

Second Place in Algebra, Mathematics Team Contest, Comillas ICAI

ADDITIONAL INFORMATION

- **Languages** | Spanish (Native) | English (Proficient; five years in London) | French (B2).
- **Technical Skills** | Python, SQL, PyTorch, scikit-learn, Hugging Face, MATLAB, R, Docker, MLOps, CI/CD, FastAPI, React, TypeScript, MongoDB, LLMs, RAG and AI Agents.
- **Teaching** | Volunteer English teacher in Trujillo, Peru (Jun-Jul 2024); sailing instructor at Escuela de Vela Socaire, Murcia (five summer seasons).